Lecture 5 (9/23)

Expectation and conditional expectation

Unit 1 Review

- Probability models: sample spaces, outcomes, events, and probability distributions
- Discrete random variables; joint, marginal, and conditional probabilities; Bayes' formula
- Independence and iid samples; simulation as an approximation to probability; translating between code and probability notation
- Continuous random variables: Uniform distribution, densities, CDFs

Goals for Unit 2:

- Expectation and variance, coefficient of variation
- Normal distribution; standardization and linear transformations
- The linear regression model assumptions
- Understand covariance and correlation

Sample averages and expectation

- The sample mean of iid draws $Y_1, \ldots, Y_N$ of a variable with sample space S is

$$\overline{Y} = \texttt{np.mean(y)} = \frac{1}{N}\sum_{i=1}^{N} Y_i = \sum_{y \in S} y \frac{\#\{Y_i = y\}}{N}$$

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- For a continuous random variable with density $f_Y(y)$

$$E[Y] = \int_{-\infty}^{\infty} y f_Y(y) dy$$

Expectation of a function

For any function h[1] of Y, we can similarly write

$$E[h(Y)] = \sum_{y \in S} h(y)\, P_Y(y) \quad \text{OR} \quad E[h(Y)] = \int h(y) f_Y(y) dy$$

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Taking $g(y) = 1_A(y)$ gives

$$E[1_A(Y)] = P(Y \in A).$$

A probability is just the expectation of an indicator.

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Question

Let Y take values in $\{1, 2, 3\}$ with $P(Y{=}1) = \frac{1}{2}$, $P(Y{=}2) = \frac{1}{3}$, $P(Y{=}3) = \frac{1}{6}$. Compute $E[Y]$ and $E[Y^2]$.

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Question

Let $Y \sim \mathrm{Uniform}(0, L)$. What is $E[Y]$? What is $E[Y^2]$?

Properties of expectation

- **Linearity:** $E[X + Y] = E[X] + E[Y]$. For a constant a, $E[aX] = aE[X]$ and $E[X + a] = E[X] + a$.
- **Factorization when independent:** If X and Y are independent, $E[XY] = E[X]E[Y]$.
- **Tower property:** $E\big[E[X \mid Y]\big] = E[X]$ where $E[X \mid Y]$ is the expectation of $X|Y$ (and hence depends on Y). This is just marginalization for expectations.

In each of the examples below, you should also understand how to write code which simulates the model and test the formula with these simulations.

Question

Let $X \sim \text{Bernoulli}(1/2)$ and suppose $Y \in \{1, 2, 3\}$ with conditional distribution

	$Y{=}1$	$Y{=}2$	$Y{=}3$
$P(Y \mid X{=}0)$	$1/2$	$1/4$	$1/4$
$P(Y \mid X{=}1)$	$1/3$	$1/3$	$1/3$

Compute $E[Y]$ and $E[XY]$.

Question

Let

$$X \sim \text{Uniform}(0, 2)$$
$$Y|X \sim \text{Uniform}(0, X)$$

Compute $E[Y]$ and $E[XY]$.

Solutions

1) $Y \in \{1,2,3\}$, $P(Y{=}1){=}\frac{1}{2}$, $P(Y{=}2){=}\frac{1}{3}$, $P(Y{=}3){=}\frac{1}{6}$: $\quad E[Y] = \frac{5}{3}$, $\;E[Y^2] = \frac{10}{3}$.
2) $Y \sim \text{Uniform}(0, L)$: $\quad E[Y] = \frac{L}{2}$, $\;E[Y^2] = \frac{L^2}{3}$.
3) $X \sim \text{Bernoulli}(\frac{1}{2})$, $Y \mid X$ from the table: $\quad E[Y] = \frac{15}{8}$, $\;E[XY] = 1$.
4) $X \sim \text{Uniform}(0, 2)$, $Y \mid X \sim \text{Uniform}(0, X)$: $\quad E[Y] = \frac{1}{2}$, $\;E[XY] = \frac{2}{3}$.